

d) $f(x) = (x+1)/(x+2)$

Solution: We know that $-1 \in \mathbb{R}$ and $1 \in \mathbb{R}$.

$$f(-1) = \frac{(-1)+1}{(-1)+2} = \frac{0}{1} = 0 \quad f(1) = \frac{1+1}{1+2} = \frac{2}{3}$$

But $-1 \neq 1$. By Defn. 5, f is not one-to-one.

By Defn. 8, f is not a bijection. $\square$

24) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and let $f(x) > 0$ for all $x \in \mathbb{R}$. Show that $f(x)$ is strictly increasing if and only if the function $g(x) = 1/f(x)$ is strictly decreasing.

Solution: Let $f(x)$ be strictly increasing. By Defn. 6, $\forall x \forall y, x < y \in \mathbb{R}$ if $x < y$, $f(x) < f(y)$. We know $f(x) > 0$ and $f(y) > 0$.

~~$\therefore 1/f(x) < 1/f(y)$~~

$\therefore 1/f(x) < 1/f(y) \Rightarrow g(x) < g(y)$. By Defn. 6, $g(x)$ is strictly decreasing. Let $g(x)$ be strictly decreasing. By Defn. 6, $\forall x \forall y, x < y \in \mathbb{R}$ if $x < y$, then $g(x) > g(y)$. $\therefore \frac{1}{f(x)} > \frac{1}{f(y)} \Rightarrow f$ (Wrong attempt)

24) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and let $f(x) > 0$ for all $x \in \mathbb{R}$. Show that $f(x)$ is strictly increasing if and only if the function $g(x) = 1/f(x)$ is strictly decreasing.

Solution: Let f be strictly increasing. By Defn. 6, $\forall x \forall y, x < y \in \mathbb{R}$ if $x < y$, $f(x) < f(y)$. We know that $f(x), f(y) \neq 0$.

$\therefore 1/f(x) > 1/f(y) \Rightarrow g(x) > g(y)$. By Defn. 6, g is strictly decreasing.

Let g be strictly decreasing. By Defn. 6, $\forall x \forall y, x < y \in \mathbb{R}$, if $x < y$, then $g(x) > g(y)$. $\therefore 1/f(x) > 1/f(y) \Rightarrow f(x) < f(y)$. By Defn. 6, f is strictly increasing. $\square$

25) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and let $f(x) > 0$ for all $x \in \mathbb{R}$. Show that $f(x)$ is strictly decreasing if and only if the function $g(x) = 1/f(x)$ is strictly increasing.

~~$\therefore 1/f(x) < 1/f(y)$~~